

DFMEA — finN-2e19 Smartphone Battery

Document: VBF-T6R1F-DFMEA-01 | **Case:** t6_r1_flash | **Date:** 2026-08-25 | **Severity/occurrence/detection:** 1–10 scale (10 = worst); RPN = S×O×D

FM-1: Lithium plating during 4C fast charge (M4)

Item	Value
Failure mode	Anode potential < 0 V at the 4.7 V charge cutoff → Li deposition on graphite surface
Effect	Capacity fade, safety risk; violates M4
Cause	LNMO OCP ceiling (4.706 V) ≈ charge cutoff (4.7 V): anode potential pinned at −0.02...−0.26 V by cathode side; $\phi_{e(\text{sep})} \approx +0.2$ V (electrolyte potential in anode)
Current controls	High-transference electrolyte ($t_{\text{Li}} \approx 0.9$, $\sigma \approx 20$ S/m, $D \approx 1\text{e-}9$) — reduces $\phi_{e(\text{sep})}$; dense short anode (120 μm , $\varepsilon \approx 0.65$) — shortens conduction path; verified at DFN
S / O / D / RPN	9 / 9 / 2 / 162
Verification	4C_charge_45C with plating model: plated (DFN −0.019 V, SPM −0.262 V) → residual risk confirmed
Recommended action	Out of DOF scope: lower charge cutoff (< 4.65 V, usage-mode — excluded), cathode with OCP ceiling > 4.7 V (electrode system locked), or asymmetric plating-aware charging profiles. Residual risk documented in design_spec §4.

FM-2: Thermal excursion at 4C (M5)

Item	Value
Failure mode	$T_{\text{max}} > 50$ °C during 4C charge at 45 °C ambient
Effect	Aging acceleration, safety margin loss
Cause	4C polarization heat + 45 °C ambient; lumped-SPM underestimates peak by 4.3 K
Current controls	Vapor chamber + graphite + Al frame: $h = 300$ W/m ² K ($h=250$ verified minimum); DFN-verified 322.5 K
S / O / D / RPN	7 / 3 / 4 / 84
Verification	DFN h-sweep: 120→326.1 fail, 200→323.8 fail, 250→323.1 pass, 300→322.5 pass
Recommended action	Keep $h \geq 300$; monitor in production with thermocouple on tab (O→2).

FM-3: SEI overgrowth > 500 nm by 100 cycles (M3)

Item	Value
Failure mode	SEI thickness exceeds 500 nm → Li inventory loss, impedance rise
Effect	Capacity fade, power loss

Item	Value
Cause	EC permeation through SEI (diffusion-limited growth $L \propto \sqrt{(D_{ec}t)}$); deep per-cycle cycling
Current controls	LiF-rich SEI (D_{ec} 2e-19 m ² /s): 292.2 nm @100 cyc (42% margin); bracket 356.6 nm @3e-19
S / O / D / RPN	6 / 2 / 5 / 60
Verification	aging_1C_100cyc (SPMe), DFN 1-cycle cross-check 27.4 nm
Recommended action	None (margin adequate); re-verify at 45 °C cycling if duty cycle changes.

FM-4: Capacity fade beyond 20% (system-level)

Item	Value
Failure mode	End-of-life capacity < 80% before warranty life
Effect	Customer-visible runtime loss
Cause	SEI consumption + (if FM-1 occurs) plating-induced loss — FM-1 is the dominant driver
Current controls	FM-1/FM-3 mitigations
S / O / D / RPN	5 / 3 / 5 / 75
Verification	aging capacity trend recorded (see log aging outputs); no contractual retention criterion
Recommended action	Tied to FM-1 resolution.

FM-5: Energy density shortfall (M1)

Item	Value
Failure mode	ED_vol < 950 Wh/L
Effect	Cannot meet product requirement
Cause	Mass/volume overrun, porosity increase, electrolyte inventory growth
Current controls	Thin separator (8 μm) / thin CC (8/6 μm); dense electrodes (ϵ_{neg} 0.65); measured 1171.7 Wh/L (23% margin)
S / O / D / RPN	4 / 2 / 4 / 32
Verification	calc-energy (contract formula), DFN
Recommended action	None.

FM-6: Plateau voltage below 4.1 V (M2)

Item	Value
Failure mode	Midpoint < 4.1 V
Effect	Fails spec; power electronics design point shifts
Cause	LNMO OCP shape (inherent) + polarization at 1C

Item	Value
Current controls	Minimal polarization (small particles r 2.0 μm , high- σ electrolyte); measured 4.145 V (DFN, margin 0.045 V)
S / O / D / RPN	5 / 1 / 4 / 20
Verification	calc-energy midpoint, DFN
Recommended action	Margin is the thinnest among passing criteria — monitor particle-size control in production.

Priority Ranking

Rank	FM	RPN	Status
1	FM-1 Plating (M4)	162	Residual risk — documented boundary
2	FM-4 Capacity fade	75	Linked to FM-1
3	FM-2 Thermal (M5)	84	Mitigated (h=300)
4	FM-3 SEI (M3)	60	Mitigated (D_ec 2e-19)
5	FM-6 Plateau (M2)	20	Passed
6	FM-5 ED (M1)	32	Passed